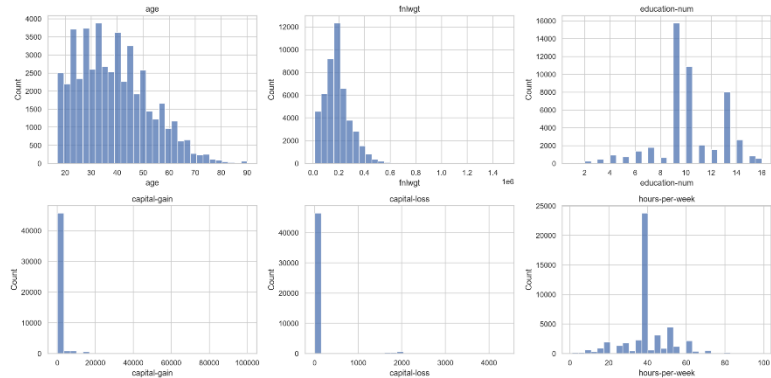


Adult Income Prediction: Problem Framing & Baselines

1. Problem Framing & Metric

Goal: Predict whether a person earns > \$50K/year to prioritize outreach (e.g., for premium products) toward likely qualified prospects. Positive class: income > \$50K. Base rate: **23.9%** of individuals earn > \$50K (11,687 of 48,842).

Primary metric: Precision (minimizes costly false positives), with **ROC AUC** as the secondary, threshold-free metric. Precision is interpreted alongside recall to ensure the model maintains useful coverage rather than achieving high precision by predicting only a few positives.



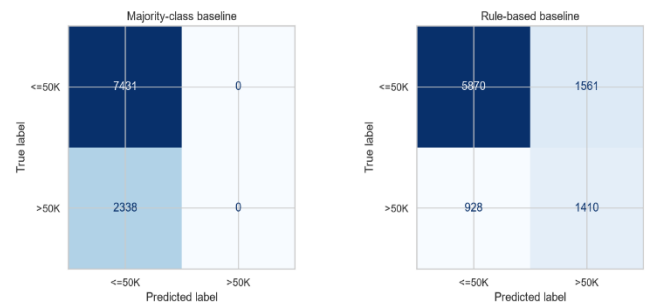
2. Reproducible Split

80% train+dev / 20% hold-out test, stratified on target, random_state=42 (train **34,189**, dev **4,884**, test **9,769** rows; ~23.9% positive rate preserved in every split). The hold-out test is touched only once for final reporting; using it during tuning would inflate reported performance.

3. Baseline Results (on hold-out test set)

model	accuracy	precision	recall	f1	roc_auc	pr_auc
Majority-class	0.7607	0	0	0	0.5	0.2393
Rule-based	0.7452	0.4746	0.6031	0.5312	0.7426	0.4946

The majority-class baseline reaches **76.1% accuracy** but **0 precision/recall** (it never predicts >50K), showing why accuracy is misleading on this imbalanced dataset. The rule-based baseline (education-num >= 13 OR capital-gain > 0) achieves **Precision = 0.4746**, **F1 = 0.5312**, and **ROC AUC = 0.743**. Since this project is evaluated primarily on **precision**, the benchmark to beat is **Precision = 0.4746**, with ROC AUC used as the secondary comparison.



4. Error Analysis & Next Steps

False negatives (missed high earners) skew toward married individuals in executive/professional occupations with high hours-per-week but no capital-gain and sub-Bachelor's education, which the rule ignores. False positives skew toward younger, highly educated people early in their careers who have not yet reached >\$50K. Missing workclass/occupation values appear in both error groups.

Primary metric going forward: Precision (secondary: ROC AUC). **Baseline to beat: Precision = 0.4746**, **ROC AUC = 0.743**.

Fixes:

1. Handle missing workclass/occupation/native-country as an explicit **Unknown** category.
2. One-hot encode categorical variables.
3. Drop the redundant string education column (keep education-num).
4. Address skew in capital-gain/capital-loss (log transform or binarize).
5. Test with/without fnlwgt (a sampling weight rather than a predictive feature).
6. Bucket rare native-country values into **Other**

